Structural Macroeconometrics

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Chapter 1

Introduction to SVAR Models

1.1 Dynamic Causal relationships in empirical macroeconomics: structural shocks and IRFs

In modern macroeconomics economists generally attempt to study the impact and propagation of structural shocks, therefore we generally find to answer questions like - (1) *What are the soruces of business cycle fluctuations?*, (2) *How does monetary policy affect the economy?*, (3) *What is the effect of an exogenous shock on macro activity/inflation?*. However we should first analyse what a **structural** in macroeconomics means.

The word **Structural** is directly linked to "Theory Oriented", "Theory driven" therefore we try to reason on the causal stands of shocks on some variables by using macroeconomic theory; thus here another important word to consider is *Causality*, since we want to find and understand what are the channels of transmission of shocks to variables.
We basically pose the following scheme:

1. **Problem of Interest**
2. **Theory/ Theories**
3. **Causal relationships** where it is possible to assess:

$$\underset{\text{Shock}}{\varepsilon_{j,t}} \xrightarrow[IRF]{} \underset{\text{Variable}}{W_{i,t+h}}$$

   where $h = 0, 1, ..., n$ and the first term relates to a structural shock of interest which is typically an unobservable latent factor, often proxied with observables and the second term is a target variable observed at time $t + h$. Now by looking at $h = 0$ we study the instantaneous impact, while $h > 0$ we observe the delayed response of the shock on the variable.

Shocks: are primitive exogenous forces that are uncorrelated with each other and should be economically meaningful.

However shocks are not directly observable so we need to find their empirical counterpart, which needs to satisfy three conditions:

1. **Endogenous** with respect to the other current and lagged endogenous variables in the model (standard OLS).

2. **Uncorrelated** with other exogenous shocks (disentangle one shock to another), admitting possible space for nonlinear dependencies, however we rule out linear correlation.

3. **Unanticipated** so agents information set at time t does not contain any shock. Having Anticipated shock might have an effect on invertibility.

To estimate the effect of a shock on a specific variable(s) over time we use the **IRF** or **Impulse Response Functions**, which enables us to trace the dynamic causal effect of the shock on the variable of interest. However also other methods exists like **Local projections**.

1.2 From small-scale monetary DSGE models to SVARs

Now we introduce this section by discussing the implications of the following example.

1.2.1 New Keynesian DSGE Model

$$\tilde{y}_t = \gamma E_t \tilde{y}_{t+1} + (1-\gamma)\tilde{y}_{t-1} - \delta(R_t - E_t\pi_{t+1}) + \zeta_{\tilde{y},t}, \tag{1.1}$$

$$\pi_t = \underbrace{\frac{\varrho}{1+\varrho\nu}}_{\beta_f} E_t\pi_{t+1} + \underbrace{\frac{\nu}{1+\varrho\nu}}_{\beta_b} \pi_{t-1} + \kappa\tilde{y}_t + \zeta_{\pi,t}, \tag{1.2}$$

$$R_t = \rho R_{t-1} + (1-\rho)\varphi_\pi(E_t\pi_{t+1} - \pi_t^*) + (1-\rho)\varphi_y\tilde{y}_t + \zeta_{R,t}. \tag{1.3}$$

The shock processes evolve as:

$$\zeta_{\tilde{y},t} = \phi_{\tilde{y}}\zeta_{\tilde{y},t-1} + \sigma_{\tilde{y}}\varepsilon_{\tilde{y},t}, \qquad \varepsilon_{\tilde{y},t} \sim WN(0,1), \tag{1.4}$$

$$\zeta_{\pi,t} = \phi_\pi\zeta_{\pi,t-1} + \sigma_\pi\varepsilon_{\pi,t}, \qquad \varepsilon_{\pi,t} \sim WN(0,1), \tag{1.5}$$

$$\zeta_{R,t} = \phi_R\zeta_{R,t-1} + \sigma_R\varepsilon_{R,t}, \qquad \varepsilon_{R,t} \sim WN(0,1). \tag{1.6}$$

The first equation represents the evolution of the output gap, defined as the difference between potential output and actual output (the *Output Gap*, i.e., the level of output in an ideal situation of perfectly functioning markets). It accounts for both past and expected future values, and also includes a component capturing the difference between the current real interest rate and expected inflation.

The second equation models the current level of inflation, which depends on several factors such as expected inflation, past inflation, and the output gap. The coefficient k is expected to be positive and measures the relationship between real economic activity and inflation.

Finally, the third equation represents the so-called *Taylor Rule*: when expected inflation exceeds the target inflation, the real interest rate R_t increases. A similar rise occurs when the output gap widens. Now we can write everything under the following compact matrix form:

$$W_t = \begin{pmatrix} \tilde{y}_t \\ \pi_t \\ R_t \end{pmatrix} = (\tilde{y}_t, \pi_t, R_t)';$$

$$M = \dim(W_t) = 3.$$

In compact matrix form:

$$\Gamma_0 W_t = \Gamma_f E_t W_{t+1} + \Gamma_b W_{t-1} + \Theta\zeta_t \tag{1.7}$$

$$\zeta_t = \Phi\zeta_{t-1} + \underbrace{\Sigma_\varepsilon^{1/2}}_{\nu_t}\varepsilon_t$$

$$E(\varepsilon_t) = 0, \quad E(\varepsilon_t\varepsilon_t') = I_M$$

where $\Gamma_0, \Gamma_f, \Gamma_b, \Theta = I_M$, $\Phi \equiv dg(\Phi)$, and $\Sigma_\varepsilon \equiv dg(\Sigma_\varepsilon)$ depend on θ. Here Φ is diagonal. Coming back to Eq.1.7 we observe that Γ_0 is the contemporaneous effect of W_t, Γ_f the forward effects of the expected variables and Γ_b is the past effect component. Our model could however hand up with multiple equilibria, this arise especially when we consider inflation before 1975 and after "Volker Rule", where volatility of inflation changes. This comes from the fact that we did allow for one degree of freedom of one variable, so controlling φ_π the central banks becomes more effective in transmitting the monetary policy.
If **a unique stable solution** exists, we can write everything in terms of a var Model, to work it out we need to proceed as follow:

- **Impose** Σ_ε to be diagonal.
- $\phi = 0$.

by doing so we obtain the following:

$$\underbrace{\begin{pmatrix} 1 & 0 & \delta \\ -k & 1 & 0 \\ (1-\rho)\varphi & 0 & 1 \end{pmatrix}}_{\Gamma_0} \begin{pmatrix} y_t \\ \pi_t \\ R_t \end{pmatrix} = \Gamma_b W_{t-1} + \Sigma_\varepsilon\varepsilon_t \ \textbf{Structural Model} \tag{1.8}$$

Now, pre-multiplying[1] by Γ_0^{-1} bot sides, we get to the following:

$$W_t = \underbrace{\Gamma_0^{-1}\Gamma_b}_{\Pi_1} W_{t-1} + \Gamma_0^{-1}\Sigma_\varepsilon\varepsilon_t, \ \textbf{Reduced Form Solution} \tag{1.9}$$

in order to work with (1.9) we need Π_1 eigenvalues to be less than 1 so $|\lambda_i| < 1$ for all i, therefore we need that the solution is unique and stable, then it can be proved that $s \in \mathrm{C}$. Given that i can write:

$$(\mathbf{I_m} - \mathbf{\Pi_1}L)W_t = \Psi\varepsilon_t$$

and therefore I can obtain[2]:

$$W_t = [\mathbf{I_m} + \mathbf{\Pi_1}^1 L(1) + \mathbf{\Pi_1}^2 L(2) + ... + \mathbf{\Pi_1}^h L(h)]U_t \tag{1.10}$$

$$= \mathbf{I_m}U_t + \mathbf{\Pi_1}^1 L(1)U_t + \mathbf{\Pi_1}^2 L(2)U_t + ... + \mathbf{\Pi_1}^h L(h)U_t \tag{1.11}$$

The following representation allows us to the effect the effects up to the past allowing for the effect to decay over time. Here Π_1 is a function of structural parameter. We therefore identify the **SVAR-B** model as:

$$W_t = \Upsilon_1 W_{t-1} + \Upsilon_2 W_{t-2} + \Psi_{\varepsilon_t}\varepsilon_t$$

The reason of the name **SVAR-B** stands from the the following: $u_t = B\varepsilon_t$, where B is the matrix we use for the functional connection of shocks to errors. We recall that the matrix Ψ is full since it maps $u_t = \Psi\epsilon_t$. As before if the unique and stable (asymptotically stationary) solution exists, it can be proved that for $s \in \mathbb{C}$:

$$\det\left(I_M - \Upsilon_1 s - \Upsilon_2 s^2\right) = 0 \quad \Longleftrightarrow \quad |s| > 1,$$

[1] This operation makes this non linear so the Π_1, Ψ depend non linearly on θ through $\Gamma_0, \Gamma_f, \Gamma_b, \Phi, \Sigma_\varepsilon$

[2] using inverse power of series $(1-x)^{-1} = \sum_0^\infty x$ for $|x| < 1$

which implies that the inverse of the matrix polynomial

$$\Upsilon(L) := I_M - \Upsilon_1 L - \Upsilon_2 L^2$$

has the representation

$$\Upsilon(L)^{-1} = \left(I_M - \Upsilon_1 L - \Upsilon_2 L^2\right)^{-1} = I_M + C_1 L + C_2 L^2 + \cdots + C_\ell L^\ell + \cdots,$$

where the sequence of matrices satisfies

$$\sum_{j=0}^{\infty} \|C_j\| < \infty, \qquad C_0 = I_M, \qquad C_\ell \xrightarrow[\ell \to \infty]{} 0,$$

(absolute summability condition). From this we can retrieve the following:

$$W_t = \sum_{j=0}^{\infty} C_j \Psi \varepsilon_t, \ \textbf{Structural VMA representation}$$

which implies that the **Impulse Response Function**:

$$IRF_{.,.}(h) : \frac{\partial W_{t+h}}{\partial \varepsilon_t'} = C_h \Psi$$

To select now a specific shock i to a variable j at h periods ahead we can propose the following:

$$IRF_{i,j}(h) := e_i' C_h \Psi e_j$$

Finally the key message here is the following:

$$\underbrace{\Upsilon(L) W_t}_{\mathbf{u_t}} = B\varepsilon_t$$

Where B should not have many zeroes however it is better to deal with many zeroes...restrictions.

1.2.2 Structural Vector Autoregressive from VAR(p) to MA(∞)

We start the entire framework by understanding how a VAR model of order p can be transformed into a Moving average and therefore catch the intertemporal dynamics of shocks. We start from a VAR(p) model:

$$Y_t = \mu + A_1 Y_{t-1} + A_2 Y_{t-2} + ... + A_p Y_{t-p} + u_t \tag{1.12}$$

The above equations is called **Reduced Form VAR**, and we notice that we need to introduce an important assumptions regarding the reduced form disturbances (white Noise Conditions):

- $E[u_t] = 0$
- $E[u_t u_t'] := \Sigma_u, \ M \times M$ pos. sem/def
- $E[u_t u_{t-k}'] = 0_{M \times M}$ this implies over time correlations and therefore makes u_t interpretable as forecast error $W_t - E(W_t | I_{t-1}) = u_t$

After stating this set of assumptions we proceed and we use the *lag operator* such that $L^1 x_t = x_{t-1}$ and write what follows:

$$\underbrace{y_t - A_1 L^1 y_t - A_2 L^2 y_t - ... - A_p L^p y_t}_{A(L)} = \underbrace{u_t}_{\Psi \varepsilon}$$

Given that we might want to isolate y_t so we need to pre multiply both sides by $A(L)^{-1}$; to do that we need $A(L)$ to be stable and therefore invertible, therefore we need that $det(A(z)) = 0 \implies det(I_m - A_1 z - A_2 z^2 - ...) = 0$ so we can change sign and divide by so we get the classic form $det(A - (\underbrace{\frac{1}{z}}_{\lambda})I_m)$ so we need $|\lambda| < 1$ inside the unit circle and therefore $z > 1$. Once the invertibility is established we can move forward and get the following **Wold representation**:

$$Y_t = (I_m + \phi_1 L^1 + \phi_2 L^2 + ...)u_t \tag{1.13}$$

here $\phi_s = \sum_{i=0}^{s} \phi_{s-i} A_i$

where ϕ_i are the dynamic multipliers. This recursion assigns the impulse–response coefficients (the "weights" of the shocks) in a way that reflects both contemporaneous interactions among variables and the dynamic, intertemporal structure of the VAR. Each future effect ϕ_s depends on the entire past propagation of shocks $(\phi_{s-1}, \phi_{s-2}, \dots)$ combined with the VAR dynamics $(A_1, A_2, \dots)$.

1.2.3 Cholesky Decomposition

As introduced in the previous sections, our objective is to recover structural shocks that satisfy the following properties:

- **Endogeneity**: shocks should affect the system contemporaneously.
- **Orthogonality**: structural shocks must be mutually uncorrelated.
- **Economic interpretability**: each shock should have a clear economic meaning.

To achieve this, we search for a linear transformation that maps the reduced–form errors into structural shocks. Formally, we seek a matrix P such that

$$u_t = P\varepsilon_t,$$

where u_t denotes the reduced-form errors and ε_t the structural shocks.

Assume that the covariance matrix of the reduced-form errors, Σ_u, is symmetric and positive definite. Since $\Sigma_u \in \mathbb{R}^{n \times n}$, it admits a decomposition of the form

$$\Sigma_u = PP',$$

where P is the Cholesky factor of Σ_u. The Cholesky decomposition yields a unique lower-triangular (or, alternatively, upper-triangular) matrix P, imposing a set of zero restrictions on the off-diagonal entries. These restrictions correspond to a recursive identification scheme that assigns a causal ordering among the variables.

Now writing our VAR in VMA we have:

$$W_t = P\varepsilon_t + C_1 P\varepsilon_{t-1} + ... + ...$$

Now reasoning on P we have that:

$$P = \frac{\partial W_t}{\partial \varepsilon_t'}$$

which represents the instantaneous response of the variables at time t. The C_i incorporates the dynamics the dynamics of the different variables, so connecting both we have that P gives the impulse of our shock while C_i replicates the dynamics of the spillover effects on the other variables. So we can identify the IRF ads follow:

$$C_h P = \frac{\partial W_t}{\partial \varepsilon_{t-h}'}$$

Analytically it is possible to recover the entries of P after estimating Σ.

$$\Sigma_u = \begin{pmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 \end{pmatrix}.$$

Assume that Σ_u admits a Cholesky factorization $\Sigma_u = PP'$, where

$$P = \begin{pmatrix} p_{11} & 0 \\ p_{21} & p_{22} \end{pmatrix}.$$

Then

$$PP' = \begin{pmatrix} p_{11} & 0 \\ p_{21} & p_{22} \end{pmatrix} \begin{pmatrix} p_{11} & p_{21} \\ 0 & p_{22} \end{pmatrix} = \begin{pmatrix} p_{11}^2 & p_{11}p_{21} \\ p_{11}p_{21} & p_{21}^2 + p_{22}^2 \end{pmatrix}.$$

Matching entries with Σ_u gives the system:

$$\begin{cases} \sigma_1^2 = p_{11}^2, \\ \sigma_{12} = p_{11}p_{21}, \\ \sigma_2^2 = p_{21}^2 + p_{22}^2. \end{cases}$$

Solving:

$$p_{11} = \sqrt{\sigma_1^2}, \qquad p_{21} = \frac{\sigma_{12}}{p_{11}}, \qquad p_{22} = \sqrt{\sigma_2^2 - p_{21}^2}.$$

Thus the Cholesky factor is

$$P = \begin{pmatrix} \sqrt{\sigma_1^2} & 0 \\ \dfrac{\sigma_{12}}{\sqrt{\sigma_1^2}} & \sqrt{\sigma_2^2 - \left(\dfrac{\sigma_{12}}{\sqrt{\sigma_1^2}}\right)^2} \end{pmatrix}.$$

Finally, for the Cholesky decomposition to exist, a *necessary condition* is that Σ_u is symmetric and positive definite.

1.3 Basic Facts on Inference of reduced form parameters

Inference is a crucial part since it enable us to test hypothesis, so in VAR model we have:

$$W_T = \mu + \Pi_1 W_{t-1} + ... + \Pi_p W_{t-p} + u_t$$

collecting $\Pi := (\Pi_1, \Pi_2, ..., \Pi_p)$ we get $M \times M(p+1)$ parameters to estimate while the covariance matrix Σ_u involves $\frac{1}{2}M(M+1)$, so we need to have enough observations to estimate our parameters. Now, writing the reduced-form parameters, we define

$$\Upsilon := (\Pi, \Sigma_u),$$

where Π collects all VAR coefficients and Σ_u is the covariance matrix of the reduced-form innovations.

To work with a vector of reduced-form parameters, we stack them as follows:

$$\delta := \left(\text{vec}(\Pi)',\ \text{vech}(\Sigma_u)'\right)' = \begin{pmatrix} \pi \\ \sigma_u^+ \end{pmatrix}.$$

The operator vec$(\cdot)$ stacks all columns of a matrix into a single column vector. The operator vech$(\cdot)$ stacks only the lower triangular part of a symmetric matrix (including the diagonal),

avoiding duplication of symmetric elements. So $\hat{\delta}_T$ can be estimated OLS and QML (Gaussian) estimator. We now consider the asymptotic distribution of the reduced-form parameter vector. Under standard regularity conditions, we have

$$T(\hat{\delta}_T - \delta_0) = \begin{pmatrix} T(\hat{\pi}_T - \pi_0) \\ T(\hat{\sigma}^+_{u,T} - \sigma^+_{u,0}) \end{pmatrix} \xrightarrow{d} N(0, V_\delta),$$

where the asymptotic covariance matrix takes the block-partitioned form

$$V_\delta := \begin{pmatrix} V_\pi & V_{\pi,\sigma_u} \\ V_{\sigma_u,\pi} & V_{\sigma_u} \end{pmatrix}.$$

If the off-diagonal element is zero we assume independence in the estimation of the two parameters, typically when we have u_t Weak White Noise then $V'_{\sigma_u,u} \neq 0$. One way to find the estimator of V_δ is through the use of bootstrap. IRFs in SVARs are nonlinear (but continuous and differentiable) functions of the reduced-form parameters δ. For each horizon $h = 0, 1, 2, \ldots$, we write

$$\phi_h = f_h(\delta),$$

where $f_h(\cdot)$ is a highly nonlinear mapping.

Consistency of $\hat{\delta}_T$ implies

$$\hat{\phi}_{h,T} = f_h(\hat{\delta}_T) \quad \xrightarrow{p} \quad f_h(\delta_0) = \phi_{h,0}.$$

By the delta method,

$$T(\hat{\phi}_{h,T} - \phi_{h,0}) \xrightarrow{d} N(0, V_{\phi_h}),$$

with

$$V_{\phi_h} := \left(\frac{\partial f_h(\delta)}{\partial \delta'}\right) V_\delta \left(\frac{\partial f_h(\delta)}{\partial \delta'}\right)'.$$

The matrix $\frac{\partial f_h(\delta)}{\partial \delta'}$ is the Jacobian of the IRF with respect to δ; computing it analytically is difficult. For this reason, V_{ϕ_h} is typically estimated by bootstrap methods using bootstrap draws of $\hat{\phi}_{h,T}$.

Chapter 2

Identification problem in structural parametric models Rothenberg 1971

In econometrics "identification" stands and reports many different meanings. Rothemberg's paper (1971), reports the "identification" problem as a matter of the possibility of drawing inferences from observed samples to an underlying theoretical structure, so **existence of a structure**.

A particular feature we use in this setting is the **Information Matrix** which measures the amount of information about an unknown parameter given we work using a parametric setting [1]. Given a likelihood function $L(\theta|x)$ then we have a function that enable use to find the optimal parameter θ given the data. To find this optimality condition we rely on the $S(\theta, x) = \frac{\partial l(\theta|x)}{\partial \theta}$. Given the score function we can work on the information matrix:

$$I(\theta) := \mathbb{E}\big[s(\theta, X)\, s(\theta, X)'\big] = \mathbb{E}\left[\left(\frac{\partial \ell(\theta \mid X)}{\partial \theta}\right)\left(\frac{\partial \ell(\theta \mid X)}{\partial \theta}\right)'\right]$$

In this case the information matrix gives us information of how the curvature of the log-likelihood function is and therefore how much information it does convey. If very steep curvature we have, since we are saying we fall very close to the maxima without moving much our parameter θ.

2.0.1 Rothenberg's Paper

We start considering realization of Y random variable as y, where Y has a generally parametric structure, and given an open set for the parameter space $A \in R^p$, so we associate $f(y, \theta)$ a specific pdf. Now the problem of distinguish between structure models relates to distinguish between parameter points, in fact given $\Theta := \{\theta_1, \theta_2, ..., \}$

DEF (1): given two parameter points (structure) θ_1 and θ_2 are said to be *observationally equivalent* if $f(y, \theta_1) = f(y, \theta_2)$ for all $y \in R^n$

DEF (2): A parameter point θ_0 in A is said to be identifiable if there is no other paramether which is observationally equivalent.

DEF: Local Identification: The point θ_0 is said to be a regular point of the information matrix $I(\theta)$ if there exists an open neighborhood of θ_0 which $I(\theta)$ has a constant rank. Therefore we need $I(\theta)$ to be non singular (checking the rank locally).

[1] Parametric models are those that are described by a specific probability distribution function and a sampling scheme. Here working with specific class of parametric pdf would

the full-rank condition is necessary and sufficient for identification. Identifiability is the minimum requirement for safe inference on θ.

For global identification instead the we have:

DEF: Global Idnetification: Let $f(y, \theta)$ be a member of an exponential function family, if $I(\theta)$ is non singular in a convex set containing A, then any θ in A is globally identifiable.

we can see that this works for VAR with Gaussian Innovations.

2.0.2 VAR with Gaussian Innovation

Given a VAR model as follow:

$$W_t = \mu + \Pi_1 W_{t-1} + ... + \Pi_p W_{t-p} + u_t$$

then we collect all the parameters in the following, $\pi = vec(\Pi')$ and we get $\sigma_u^+ = vech(\Sigma_u)$

so in total we have π having $g = M + M^2 p$ and for σ_u^+ we get $m = 1/2M(M+1)$. To further see this, it can be proved that in the VAR with Gaussian innovations above, the Information Matrix reads:

$$I_T(\delta) = T \begin{pmatrix} \left(\Sigma_u^{-1} \otimes V_{XX}\right) & 0_{g \times m} \\ 0_{m \times g} & \frac{1}{2} D_M' \left(\Sigma_u^{-1} \otimes \Sigma_u^{-1}\right) D_M \end{pmatrix}$$

where $V_{XX} := \mathbb{E}(X_t X_t') < \infty$ is $k \times k$ positive definite (hence nonsingular) and D_M is the duplication matrix, i.e. the $M^2 \times m$ matrix (of full column rank m) that solves:

$$D_M \operatorname{vech}(\Sigma_u) = \operatorname{vec}(\Sigma_u).$$

It turns out that:

$$\operatorname{rank}[I_T(\delta)] = \operatorname{rank}[\Sigma_u^{-1} \otimes V_{XX}] + \operatorname{rank}\left[\frac{1}{2} D_M' \left(\Sigma_u^{-1} \otimes \Sigma_u^{-1}\right) D_M\right] = g + m = f$$

and this result is valid almost everywhere in the parameter space of σ_u^+ (i.e. the space of models with positive definite covariance matrix).

2.1 Structure and Reduced Form

In many situation we do not have direct knowledge of $f(y, \theta)$, instead we might know that Y depends on the p structural parameters in θ only through a set of reduced r parameters, so we know there exists a mapping:

$$\psi = g(\theta)$$

where g is a continuous differentiable function, so we know $f^*(y, \psi)$, so we have:

$$f^*(y, \psi) = f^*(y, g(\theta)) = f(y, \theta)$$

Now the question of our interest is to reverse the mapping in order to recover the structural parameter vector θ. In other words, we observe the reduced-form object $g(\alpha)$ and we want to know whether, at least locally, it uniquely determines α. To justify this "inversion" we can rely on the **Implicit Function Theorem**.

Intuition. The Implicit Function Theorem tells us when we can "solve" an equation of the form

$$g(\theta) = \psi$$

for
theta as a function of ψ, even if no closed-form inverse exists. Geometrically, the theorem states that if the Jacobian matrix $J($
theta) has full column rank at a point
$theta^0$, then around that point the mapping behaves like a smooth, well-behaved surface that can be locally inverted.

This ensures that small changes in the reduced form ψ correspond to unique, smooth changes in the structural parameter
theta. Hence, checking that $J($
$theta^0$) has rank p provides the key condition for *local identifiability.*

So we need a **Necessary Condition** for identification so $r \geq p$.

2.2 From Structural to Identification

We start from a simple demand/supply model for a good:

$$q_t = \gamma_{1,1} p_t + \lambda_{1,1} q_{t-1} + \lambda_{1,2} p_{t-1} + \sigma_d \varepsilon_{d,t}, \tag{2.1}$$
$$p_t = \gamma_{2,1} q_t + \lambda_{2,2} p_{t-1} + \sigma_s \varepsilon_{s,t}, \qquad t = 1, \ldots, T, \tag{2.2}$$

where:

- q_t is the quantity of the good at time t (q_0 given);
- p_t is the price of the good at time t (p_0 given);
- ε_t^d and ε_t^s are demand and supply shocks at time t, respectively, both assumed for simplicity i.i.d. $N(0,1)$;
- $\gamma_{1,1} < 0$, $\gamma_{2,2} > 0$, $\lambda_{1,1}, \lambda_{1,2}, \lambda_{2,1}, \lambda_{2,2} > 0$, $\sigma_d > 0$, and $\sigma_s > 0$ are parameters that we call *structural parameters.*

As we can see here we have a structural model which involves contemporaneous relations, i.e relations between different variables at time t. Given this model we can set the θ as containing all the parameters which involves $dim(\theta) = 7$. We can rewrite our system as follow:

$$\underbrace{\begin{pmatrix} 1 & -\gamma_{1,1} \\ -\gamma_{2,2} & 1 \end{pmatrix}}_{A} \underbrace{\begin{pmatrix} q_t \\ p_t \end{pmatrix}}_{Y_t} = \underbrace{\begin{pmatrix} \lambda_{1,1} & \lambda_{1,2} \\ 0 & \lambda_{2,2} \end{pmatrix}}_{\Gamma} \underbrace{\begin{pmatrix} q_{t-1} \\ p_{t-1} \end{pmatrix}}_{Y_{t-1}} + \underbrace{\begin{pmatrix} \sigma_d & 0 \\ 0 & \sigma_s \end{pmatrix}}_{\Omega_\varepsilon} \underbrace{\begin{pmatrix} \varepsilon_{d,t} \\ \varepsilon_{s,t} \end{pmatrix}}_{\varepsilon_t}.$$

So we can write this as:

$$AY_t = \Gamma Y_{t-1} + \Omega_\varepsilon^{1/2} \varepsilon_t$$

In this way we are able to establish a structural framework catching the simultaneous relations between the variables. Now this model is derived from the **Theory**, but since we deal with data we would ask *Is there any way to recover the structural parameters from the data?* So basically can we retrieve the parameters in a consistent way?

A first step we can set to address such question would be to get the reduced form:

$$Y_t = \underbrace{A^{-1}\Gamma}_{\Pi_1} Y_{t-1} + \underbrace{A^{-1}\Omega_\varepsilon^{1/2}\varepsilon_t}_{u_t}$$

This model is a VAR and can be estimated however the parameters that we find are not the ones we are looking for from the structural model. However this parameters maps the structural parameters given that the number of the number of parameters of the structural model are at most as the number of parameters of the reduced form.

So if we define:

$$\delta = \begin{pmatrix} vec(\Pi_1 \\ vech(\Sigma_u) \end{pmatrix} = g(\theta)$$

As we have seen from the previous section a condition needed to consistently get the structural parameters would be that:

$$|\delta| > |\theta|$$

given that this condition is meet we can use the **Implicit function Theorem** which states that $\delta \in R^q, \theta \in R^p : (\theta_0, \delta_0) \in \mathcal{A} \implies F(\delta_0, \theta_0) = 0_{p\times 1}$

$$\begin{pmatrix} vec(A\Pi_1 - \Gamma & = vec(0) \\ vech(A\Sigma_u A^{-1}) & = vec(0) \end{pmatrix} = F(\delta, \theta)$$

Given this then:

$$det = \left[\frac{\partial F(\delta_0, \theta_0)}{\partial \theta'} \right] \neq 0$$

Now working locally we can obtain that:

$$\frac{\partial F(\delta_0, \theta_0)}{\partial \theta'} = \frac{\partial g(\theta)}{\partial \theta'} =: J(\theta)_{q\times p}$$

Locally, the structural parameters and the reduced-form parameters encode the same variation, because the mapping between them is locally invertible.

Chapter 3

B-Model and Identification

In this chapter we discuss the B-Model and all implications regarding identifications schemes. The **B Model** is generally described by the following structure:

$$u_t = B\varepsilon_t$$

where the shocks are assumed to have $E[\varepsilon_r] = 0$ and $E[\varepsilon_t \varepsilon_t'] = I_{M\times M}$. We can think of B as a function mapping shocks into the innovations where B also refer to the instantaneous impact of a shock on other variables and it contains the structural parameters.

Given this setting we can write the variance covariance matrix of the innovations as:

$$\Sigma_u = B\varepsilon_t(B\varepsilon_t)' = BB'$$

Such matrix has dimension M^2 however we have that we only have $\frac{1}{2}M(M+1)$ free parameters. Given Σ_u we need to impose some conditions such that also B can be identified, in particular B has dimension $M \times M$. Let now identify $B = B(\beta)$ to indicate the non zero and free elements of B so we get:

$$\underbrace{\text{vech}(\Sigma_u)}_{\sigma_u^+} = \underbrace{\text{vech}(BB')}_{g(\beta)}.$$

where $g(\beta)$ is a nonlinear but continuous and differentiable function. What we need now from the previous chapter (Rothemberg) are the two conditions:

- **Necessary conditions**: number of free elements should be $p < \frac{1}{2}M(M+1)$ since we cannot find more elements of B given a limited number of elements of Σ_u.
- **Necessary and Sufficient condition**: we need here that $rank[J_B(\beta)] = p$ in a neighborhood of β_0 where $J_B(\beta) := \frac{\partial g(\beta)}{\partial \beta'}$ which is $m \times p$

To compute the Jacobian $J_B(\beta) = \partial g(\beta)/\partial\beta'$, we adopt the explicit-form representation of the linear restrictions on B:

$$\text{vec}(B) = S_B\beta + s_B,$$

where S_B is an $M^2 \times p$ selection matrix collecting the free elements of B, and s_B is an $M^2 \times 1$ vector containing the fixed (known) elements.

By applying standard matrix differential calculus, the Jacobian $J_B(\beta)$ of $g(\beta) = \text{vech}(BB')$ with respect to the free parameters β is

$$J_B(\beta) = \frac{\partial \sigma_u^+}{\partial \beta'} = \frac{\partial g(\beta)}{\partial \beta'} = \frac{\partial \,\text{vech}(B(\beta)B(\beta)')}{\partial \beta'}.$$

Using the chain rule,

$$J_B(\beta) = \frac{\partial \,\text{vech}(BB')}{\partial \,\text{vec}(B)'}\frac{\partial \,\text{vec}(B)}{\partial \beta'} = D_M^+ \frac{\partial \,\text{vec}(BB')}{\partial \,\text{vec}(B)'} S_B.$$

Since

$$\frac{\partial \operatorname{vec}(BB')}{\partial \operatorname{vec}(B)'} = 2N_M(B \otimes I_M),$$

we obtain

$$J_B(\beta) = D_M^+\{2N_M(B \otimes I_M)\}S_B = 2D_M^+(B \otimes I_M)S_B,$$

where $D_M^+ = (D_M'D_M)^{-1}D_M'$ is the Moore–Penrose pseudoinverse of the duplication matrix D_M and N_M is the symmetrizer matrix.

Exactly Identified vs Overidentified

Whenever we consider a **B model** we saw we need to impose restrictions to have identifiability. In such setting nothing prevent to put more restrctions tha the minimum required by the necessary condition discussed above. Therefore here we discuss:

- **Exact identification**: if the necessary and sufficient condition is satisfied and teh necessary order condition is satisfied with $b = \frac{1}{2}M(M+1)$
- **Overidentified**: we have the same as above but $b < \frac{1}{2}M(M+1)$, allowing also for testing such restrictions using appropriate testing procedures (Wald

To check the necessary and sufficient rank condition for identification[1] we can either:

- Generate randomly many values of $\tilde{\beta}$ of β from a specific distribution and check whether the necessary and sufficient condition hold.
- Estimate the SVAR by Maximum Likelihood and given $\hat{B}_t = B(\hat{\beta}_T)$ and then check the necessary and sufficient condition.[2]

3.1 Resizing and rescaling Structural Shocks

Many times, when computing impulse response functions, we are interested in generating a shock of a specific magnitude in order to simulate the real effects of an economic policy intervention. To impose such a shock, we consider

$$B\delta,$$

where $\delta = (\delta_1, 0, 0)'$ and δ_1 represents the size of the structural shock $\varepsilon_{1,t}$.

From the system we obtain:

$$B\delta = \begin{pmatrix} \beta_{11} & 0 & \beta_{13} \\ \beta_{21} & \beta_{22} & 0 \\ 0 & \beta_{32} & \beta_{33} \end{pmatrix} \begin{pmatrix} \delta_1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} \beta_{11}\delta_1 \\ \beta_{21}\delta_1 \\ 0 \end{pmatrix}.$$

If we want the impact on the second variable to be equal to a desired magnitude x, then we must impose:

$$\beta_{21}\delta_1 = x \qquad \Longrightarrow \qquad \delta_1 = \frac{x}{\beta_{21}}.$$

Here x is the size of the shock we want to simulate on the other variables.

[1] Identification holds up to sign normalization, recall that if i cange one sign then i have to change the sign of the entire column

[2] If likelihood is flat than we have infinite values of the parameter that are mapped to the same value of the likelihood.

Chapter 4

AB Model

The AB model is the most general model within the structural framework, due to its structure it can be transformed either in an A-Model or in a B-Model. We identify the model as follo:

$$AW_t = \tau + \Upsilon_1 W_{t-1} + ... + \Upsilon_p W_{t-p} + Au_t$$

we can then identify the following:

$$Au_t = B\varepsilon_t$$

and if A is invertible we can obtain:

$$u_t = \underbrace{A^{-1}B\varepsilon_t}_{C}$$

This model is more flexible because the matrix A allows us to incorporate general simultaneous interactions among the endogenous variables, while the matrix B governs the instantaneous effect of structural shocks on the reduced-form residuals Au_t.

However, the matrix that captures the *true* instantaneous impact of each structural shock on the reduced-form errors u_t is

$$C = A^{-1}B.$$

The matrix C embodies both the instantaneous effect of the shocks and the contemporaneous interdependence among the variables encoded in A^{-1}.

Therefore, even if a structural shock directly affects only a single variable (i.e., only one element of the corresponding column of B is nonzero), the shock can still propagate contemporaneously to the other variables through the simultaneous linkages contained in A^{-1}.

- **Pros**: few restrictions if zero at all.
- **Cons**: we need to specify both A and B such that the model is identified.

From the implied moment condition we get:

$$A\Sigma_u A' = BB' \implies \Sigma_u = A^{-1}BB'(A')^{-1}$$

given that we recall that our Σ_u has $1/2M(M+1)$ free elements it places such restriction on the $2M^2$ elements of A and B, therefore we need $2M^2 - 1/2M(M+1)$ restrictions to be imposed to achieve identifiability[1].

[1]Note that for $A = I_M$ the models collapses to a B model and vice-versa

Linear restrictions

Consider the structural matrices A and B. We impose linear restrictions by parameterising their vectorised forms as

$$\text{vec}(A) = S_A\,\alpha + s_A, \tag{4.1}$$
$$\text{vec}(B) = S_B\,\beta + s_B, \tag{4.2}$$

where:

- S_A is an $M^2 \times b_A$ selection matrix and α is a $b_A \times 1$ vector collecting the unrestricted entries of A;
- s_A is an $M^2 \times 1$ vector containing the fixed (known) elements of A;
- S_B is an $M^2 \times b_B$ selection matrix and β is a $b_B \times 1$ vector containing the free parameters in B;
- s_B is an $M^2 \times 1$ vector with the pre-specified elements of B.

A more compact representation is obtained by stacking the restrictions:

$$\begin{pmatrix}\text{vec}(A)\\ \text{vec}(B)\end{pmatrix} = \begin{pmatrix}S_A & 0\\ 0 & S_B\end{pmatrix}\begin{pmatrix}\alpha\\ \beta\end{pmatrix} + \begin{pmatrix}s_A\\ s_B\end{pmatrix}.$$

This shows that the full vector of unrestricted structural coefficients in the AB-model is

$$\gamma = \begin{pmatrix}\alpha\\ \beta\end{pmatrix}, \qquad \dim(\gamma) = b_A + b_B = b.$$

Order condition for identification. A necessary requirement for identification is that the number of free structural parameters does not exceed the number of available moment conditions implied by the symmetric reduced-form covariance matrix. Hence,

$$b = b_A + b_B \;\le\; m = \frac{1}{2}M(M+1).$$

Equivalently, in terms of restrictions, letting l_A and l_B denote the number of constraints imposed on A and B respectively, the condition becomes

$$l = l_A + l_B \;\ge\; 2M^2 - \frac{1}{2}M(M+1).$$

Once we have defined a necessary condition we need to introduce also a necessary and sufficient rank condition such as:

$$\Sigma_u = CC'$$

and

$$\sigma_u^+ = g(\gamma) = vech[A^{-1}BB'(A')^{-1}]$$

so we can focus on the Jacobian matrix and get:

$$J_{A,B}(\gamma) = \frac{\partial \sigma_u^+}{\partial \gamma'} = \left(\frac{\partial \sigma_u^+}{\partial \alpha'}, \frac{\partial \sigma_u^+}{\partial \beta'}\right)$$

Jacobian and rank condition for identification

By direct matrix algebra, the Jacobian of the mapping from the unrestricted parameters γ to the implied covariance restrictions is

$$J_{A,B}(\gamma) = 2D_M^+\Big(-(\Sigma_u^{AB} \otimes A^{-1}) \;:\; (A^{-1}B \otimes A^{-1})\Big) \begin{pmatrix} S_A & 0 \\ 0 & S_B \end{pmatrix},$$

where

$$\Sigma_u^{AB} = A^{-1}BB'(A')^{-1}$$

is the covariance matrix of the reduced-form innovations under the AB-model restrictions.

A necessary and sufficient condition ensuring local identification of the structural parameters is that the Jacobian evaluated at the true parameter vector γ_0 has full column rank:

$$\operatorname{rank}(J_{A,B}(\gamma_0)) = b_A + b_B.$$

Summary: identification of the AB-model

Consider the structural system

$$Au_t = B\varepsilon_t, \qquad \Sigma_u = A^{-1}BB'(A')^{-1},$$

and impose

$$l = l_A + l_B = M^2 - b$$

linear restrictions on the elements of A and B. The restrictions can be written as

$$\begin{pmatrix} \operatorname{vec}(A) \\ \operatorname{vec}(B) \end{pmatrix} = \begin{pmatrix} S_A & 0 \\ 0 & S_B \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} + \begin{pmatrix} s_A \\ s_B \end{pmatrix},$$

where α and β collect the $b = b_A + b_B$ free structural parameters.

The necessary order condition for identification is

$$b = b_A + b_B \;\le\; \frac{1}{2}M(M+1),$$

which is equivalent to requiring a sufficiently large number of restrictions:

$$l = l_A + l_B \;\ge\; 2M^2 - \frac{1}{2}M(M+1).$$

Estimation of A and B

Starting from the concentrated log-likelihood (see Appendix), the SVAR restrictions may be written as $\sigma_{u,t}^+ = g(\gamma)$, so that

$$\ln L(g(\gamma)) = C - \frac{T}{2}\log\big[\det\big(A^{-1}BB'(A')^{-1}\big)\big] - \frac{T}{2}\operatorname{tr}\Big[A'B^{-1}B^{-1}A\,\hat{\Sigma}_u\Big].$$

The maximum likelihood estimator of the structural parameter vector γ is defined as

$$\hat{\gamma}_{T,ML} \;=\; \arg\max_{\gamma}\; \ln L_T(g(\gamma)),$$

and is obtained numerically.

Under standard regularity conditions for stationary VARs, the ML estimator $\hat{\gamma}_{T,ML} \equiv \hat{\gamma}_T$ is consistent and asymptotically normal:

$$\hat{\gamma}_T \xrightarrow{p} \gamma_0, \qquad T^{1/2}(\hat{\gamma}_T - \gamma_0) \xrightarrow{d} N(0_{b\times 1}, V_\gamma),$$

where

$$V_\gamma = (I_\infty(\gamma_0))^{-1}.$$

Asymptotic covariance and standard errors

From classical ML theory,

$$\hat{V}_\gamma = \left[-\frac{1}{T} H_T(\hat{\gamma}_T)\right]^{-1} \xrightarrow{p} \left(I_\infty(\gamma_0)\right)^{-1},$$

where

$$H_T(\hat{\gamma}_T) = \left.\frac{\partial^2 \ln L_T(\gamma)}{\partial\gamma\,\partial\gamma'}\right|_{\gamma=\hat{\gamma}_T}.$$

Software packages that compute SVARs usually report standard errors equal to the square roots of the diagonal entries of $[-H_T(\hat{\gamma}_T)]^{-1}$.

Information matrix from the AB-model

From the identification theory of the AB-model,

$$I_T(\gamma) = J_{A,B}(\gamma)'\, I_T(\sigma_u^+)\, J_{A,B}(\gamma),$$

where $I_T(\sigma_u^+)$ is the $m \times m$ information matrix associated with the reduced-form covariance parameters.

Consequently,

$$I_\infty(\gamma) = \lim_{T\to\infty} \frac{1}{T} I_T(\gamma) = J_{A,B}(\gamma)'\, I_\infty(\sigma_u^+)\, J_{A,B}(\gamma),$$

and a consistent estimator is

$$\widehat{I}_\infty(\hat{\gamma}_T) = J_{A,B}(\hat{\gamma}_T)'\, I_\infty(\hat{\sigma}_{u,T}^+)\, J_{A,B}(\hat{\gamma}_T).$$

Thus,

$$\hat{V}_\gamma = \left(I_\infty(\hat{\gamma}_T)\right)^{-1}.$$

*Recovering $\hat{A}_T$ and $\hat{B}_T$

Let $\hat{\gamma}_T = (\hat{\alpha}_T', \hat{\beta}_T')'$. The estimated restricted matrices are obtained from

$$\mathrm{vec}(\hat{A}_T) = S_A \hat{\alpha}_T + s_A, \qquad \mathrm{vec}(\hat{B}_T) = S_B \hat{\beta}_T + s_B.$$

Hence, $\hat{A}_T = A(\hat{\alpha}_T)$ is obtained by reshaping $\mathrm{vec}(\hat{A}_T)$, and similarly $\hat{B}_T = B(\hat{\beta}_T)$.

Structural IRFs

Structural impulse responses are defined as

$$\mathrm{IRF}_{\bullet,\bullet}(h) = \Phi_h A^{-1} B, \qquad h = 0, 1, 2, \ldots.$$

Given the estimates $(\hat{A}_T, \hat{B}_T)$ and $\hat{\Phi}_{h,T}$, the empirical IRFs are computed as

$$\widehat{\mathrm{IRF}}_{\bullet,\bullet}(h) = \hat{\Phi}_{h,T}\, \hat{A}_T^{-1} \hat{B}_T, \qquad h = 0, 1, 2, \ldots.$$

4.1 Blanchard and Perotti's Fiscal model

Blanchard and Perotti (2002) (hereafter BP) estimate a structural VAR of the AB-type in order to quantify the size of fiscal multipliers in the United States. Their key objective is to measure how output reacts to unexpected changes in government spending and tax revenues.

The idea is straightforward: once we have a VAR for output, taxes, and public expenditures, we can study how each variable responds to exogenous fiscal shocks. The main challenge is to

isolate *exogenous* changes in fiscal policy from those that are simply automatic responses to the business cycle. The AB-SVAR framework provides a clean way to impose these identification restrictions.

BP consider a three-variable system ($M = 3$):

$$W_t = \begin{pmatrix} \log(\text{Tax}_t) \\ \log(G_t) \\ \log(\text{GDP}_t) \end{pmatrix},$$

where the first component is the log of total tax revenues, the second is log government spending, and the third is log output. The sample consists of quarterly US data from 1950Q1 to 2006Q4 ($T = 228$).

Reduced-form specification

After selecting the appropriate lag length, the reduced-form VAR takes the form

$$W_t = \Pi_1 W_{t-1} + \Pi_2 W_{t-2} + \Pi_3 W_{t-3} + \Pi_4 W_{t-4} + \mu + \delta t + \lambda D_t^{75Q2} + u_t.$$

The deterministic part of the VAR consists of:

- a constant (μ),
- a linear trend (δt) capturing long-term growth,
- a dummy D_t^{75Q2} equal to 1 in 1975Q2 (and 0 otherwise) to absorb a major temporary tax event (a "well-identified", isolated tax cut).

AB-form representation

Given the reduced form, the structural representation is

$$AW_t = A\mu + A\delta t + A\lambda D_t^{75Q2} + A\Pi_1 W_{t-1} + \cdots + A\Pi_4 W_{t-4} + B\varepsilon_t,$$

where the A matrix captures the contemporaneous interactions among the variables, while B links structural shocks to the system.

Because $M = 3$, the AB-model requires at least

$$l = l_A + l_B \geq 2M^2 - \tfrac{1}{2}M(M+1) = 12$$

identifying restrictions on A and B.

Intuitively, these restrictions tell us which variables can contemporaneously affect which others, and which fiscal shocks can affect output within the same quarter.

Baseline BP identification

BP impose the following structure for A:

$$A = \begin{pmatrix} 1 & 0 & -0.8 \\ 0 & 1 & 0 \\ a_{31} & a_{32} & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} b_{11} & b_{12} & 0 \\ 0 & b_{22} & 0 \\ 0 & 0 & b_{33} \end{pmatrix}, \qquad u_t = B\varepsilon_t.$$

Meaning of the restrictions:

- The element $a_{13} = -0.8$ is calibrated rather than estimated. It reflects the elasticity of tax revenues with respect to current output. Taxes respond contemporaneously to the business cycle (automatic stabilizers).
- The element $a_{12} = 0$ imposes that government spending does not respond instantaneously to tax revenues. This formalizes the idea that spending is subject to institutional rigidities.
- Either $b_{12} = 0$ (spending shocks do not affect taxes contemporaneously) or $b_{21} = 0$ (tax shocks do not affect spending contemporaneously) can be imposed. BP choose $b_{21} = 0$, and leave b_{12} unrestricted.

These restrictions reflect institutional timing assumptions:

- Taxes can respond quickly to economic conditions.
- Government spending cannot adjust instantly.
- Fiscal authorities do not typically revise spending plans within the quarter in response to unexpected tax changes.

Together, these restrictions produce the required $l = 12$ constraints that identify the SVAR.

Estimation and computation of IRFs

The estimation yields $\hat{A}_T$ and $\hat{B}_T$, obtained via

$$\text{vec}(\hat{A}_T) = S_A\hat{\alpha}_T + s_A, \qquad \text{vec}(\hat{B}_T) = S_B\hat{\beta}_T + s_B.$$

The structural impulse responses follow from

$$\text{IRF}(h) = \Phi_h A^{-1} B, \qquad h = 0, 1, 2, \ldots,$$

and their empirical counterparts are

$$\widehat{\text{IRF}}(h) = \hat{\Phi}_{h,T}\hat{A}_T^{-1}\hat{B}_T.$$

These IRFs measure how fiscal shocks propagate: tax shocks, spending shocks, and output shocks are each traced through the dynamic system.

In BP's context, the IRFs display the fiscal multipliers: they show by how much GDP moves in response to a one-unit exogenous change in government spending or taxes, taking into account all propagation channels embedded in the SVAR.

Chapter 5

Confidence intervals for IRF

So far, we have seen how to compute IRFs; in particular, we have focused on how to find the point estimate of the IRFs. However, we did not look at the uncertainty surrounding our estimates; in fact, this is crucial to assess and test whether some particular IRFs might or not be defined as statistically significant.

5.1 Analytic Confidence Bands for IRF

In the most general case, we can define the IRF as follows:

$$IRF_{\cdot,\cdot}(h) = \phi_h = C_h K$$

This identifies the dynamic causal effect of shocks on variables. In the previous section, we have seen that $C_h = R(\mathcal{C}^{(})\mathcal{R}'$ and K is dependent on the type of structural SVAR we want to build, hence $K = P$ for cholesky, $K = B$ for the B model and so on. The IRF we have identified is attain to our model however we need an estimator for it, thus through the plug-in method we obtain:

$$\widehat{IRF}_{\cdot,\cdot}(h) = \hat{\phi}_h = \hat{C}_h \hat{K}$$

To obtain the effect of a specific shock from variable j to variable i, after h lags, we use:

$$\phi_{i,j,h} = e_i' C_{h,T} K e_j$$

Using the plug-in estimator we can obtain that

$$\hat{\phi}_{i,j,h} = f(\hat{\rho}_T),$$

and to study its uncertainty we need the joint asymptotic distribution of all elements of $\rho = (\pi', \gamma')'$. To this end, recall that under the assumptions $u_t \sim iid\, N(0, \Sigma_u)$ for $t = 1, \ldots, T$ (or, more generally, u_t forming an MDS with conditional homoskedasticity), the ML estimators of the reduced-form parameters of the VAR satisfy

$$T^{1/2}(\hat{\delta}_T - \delta_0) = T^{1/2} \begin{pmatrix} \hat{\pi}_T - \pi_0 \\ \hat{\sigma}^+_{u,T} - \sigma^+_{u,0} \end{pmatrix} N\left(\begin{pmatrix} 0_{g\times 1} \\ 0_{m\times 1} \end{pmatrix}, \begin{pmatrix} V_\pi & 0_{g\times m} \\ 0_{m\times g} & V_{\sigma^+_u} \end{pmatrix} \right),$$

where

$$V_\pi = (\Sigma_u \otimes V_{XX}^{-1}), \qquad V_{XX} = E(X_t X_t'),$$

and

$$V_{\sigma^+_u} = 2D^+_M(\Sigma_u \otimes \Sigma_u)(D^+_M)'.$$

These results provide the asymptotic distribution of the reduced-form parameters (π, σ_u^+), which is the key ingredient for deriving the asymptotic distribution of the impulse response estimator $f(\hat{\rho}_T)$ via the delta method. So finally we get:

$$\sqrt{T}(\hat{\rho}_T - \rho_0) \xrightarrow[T\to\infty]{d} N(0_{g+b)\times 1}, V_\rho)$$

5.1.1 Deltha Method

So far, we have seen that the $\sqrt{T}(\hat{\rho}_T - \rho_0)$ is normally distributed; however, this does not imply that the IRF is also normally distributed, since the IRF is given by a nonlinear function of ρ.

We can show this by using the mean value theorem which gives us:

$$\hat{\phi} = f(\rho_0) + \frac{\partial f(\hat{\rho_T} - \rho_0)}{\partial \theta'} + op(1)$$

Therefore we get:

$$\sqrt{T}(\hat{\phi_T} - \phi_0) = J(\rho_0)\sqrt{T}(\hat{\rho_T} - \rho_0)$$

Let $\hat{\rho}_T$ be a consistent and asymptotically normal estimator,

$$T^{1/2}(\hat{\rho}_T - \rho_0) \xrightarrow[T\to\infty]{d} N(0, V_\rho),$$

and let $\hat{\phi}_T = f(\hat{\rho}_T)$ for some continuously differentiable mapping $f : \mathbb{R}^a \to \mathbb{R}^d$ with $f(\rho_0) = \phi_0$. By the delta method,

$$T^{1/2}(\hat{\phi}_T - \phi_0) \xrightarrow[T\to\infty]{d} N(0, V_\phi), \qquad V_\phi = G_\rho V_\rho G_\rho',$$

where

$$G_\rho = \frac{\partial f(\rho_0)}{\partial \rho'}$$

is the $d \times a$ Jacobian.

Applying this to the IRF estimator $\hat{\phi}_{i,j,h} = f(\hat{\rho}_T)$, with $\rho = (\pi', \gamma')'$ and $\dim(\rho) = g + b$, yields

$$T^{1/2}(\hat{\phi}_{i,j,h} - \phi^0_{i,j,h})N(0, V_{\phi_{i,j,h}}), \qquad V_{\phi_{i,j,h}} = G_\rho V_\rho G_\rho',$$

with G_ρ now a $1 \times (g + b)$ gradient.

In practice G_ρ can be evaluated numerically, and a consistent estimator of $V_{\phi_{i,j,h}}$ is

$$\widehat{V}_{\phi_{i,j,h}} = G_{\hat{\rho}_T}\, \hat{V}_\rho\, G'_{\hat{\rho}_T}.$$

The corresponding standard error is

$$\text{s. e.}(\hat{\phi}_{i,j,h}) = \left(\frac{1}{T}\widehat{V}_{\phi_{i,j,h}}\right)^{1/2}.$$

Thus, a $(1 - \zeta)$ confidence interval follows from asymptotic normality:

$$\Pr\left(q_\zeta \le \frac{\hat{\phi}_{i,j,h} - \phi^0_{i,j,h}}{\text{s. e.}(\hat{\phi}_{i,j,h})} \le q_{1-\zeta}\right) = 1 - \zeta,$$

where q_ζ and $q_{1-\zeta}$ denote standard normal quantiles.

5.2 Bootstrap Methods

So far, we have seen how it is possible to retrieve the distribution of the IRFs analytically to compute the confidence bands of the IRF; however, Bootstrap methods are widely spread and more used in standard applications, why?

- **Normal approximation works when $T \to \infty$, not when we have a small finite sample**
- **Computational convenience and better approximation in finite sample**

The idea of **Bootstrap** is to treat, in our case, the VAR parameters as pseudo true-population values. The model is then used to generate *pseudo samples*. Assuming a VAR(p) model with innovations being i.i.d. and normally distributed $u_t \sim i.i.d. N(0_{M\times 1}, \Sigma_u)$, we can use the **Residual-based parametric i.i.d bootstrap** which relies on the parametric known distributions of the innovations:

Residual–based parametric iid bootstrap
Given the ML (OLS) estimates $\hat{\mu}$, $\hat{\Pi}_1, \ldots, \hat{\Pi}_p$ and the residual covariance matrix $\hat{\Sigma}_u$ from the sample, the residual–based parametric iid bootstrap proceeds as follows:

1. Generate bootstrap innovations. Draw $u_t^* \sim iid\ N(0, \hat{\Sigma}_u)$ for $t = 1, \ldots, T$.

2. Generate the bootstrap sample $\{W_t^*\}_{t=1}^T$. Using the estimated VAR(p) model,

$$W_t^* = \hat{\mu} + \hat{\Pi}_1 W_{t-1}^* + \cdots + \hat{\Pi}_p W_{t-p}^* + u_t^*, \qquad t = 1, \ldots, T,$$

where the initial conditions $W_0, \ldots, W_{1-p}$ are fixed at their original sample values.

3. Re-estimate the SVAR on the bootstrap data. Estimate the chosen structural VAR model (B-model, A-model, or AB-model) using $\{W_t^*\}$ and obtain the bootstrap estimates $\hat{\mu}^*$, $\hat{\Pi}_1^*, \ldots, \hat{\Pi}_p^*$ and $\hat{\Sigma}_u^*$. From these, compute the VMA coefficients $\hat{C}_h^*$, the structural parameters $\hat{K}^*$, and the bootstrap IRF

$$\hat{\phi}_{i,j,h}^* = \hat{C}_h^* \hat{K}^*.$$

4. Repeat the procedure. Repeat steps 1–3 a total of N times to obtain

$$\hat{\phi}_{i,j,h}^{*(1)},\ \hat{\phi}_{i,j,h}^{*(2)},\ \ldots,\ \hat{\phi}_{i,j,h}^{*(N)}.$$

5. Construct bootstrap confidence intervals. Let $q_{\zeta/2}^*$ and $q_{1-\zeta/2}^*$ be the empirical $\zeta/2$ and $1-\zeta/2$ quantiles of $\{\hat{\phi}_{i,j,h}^{*(b)}\}_{b=1}^N$.

- *Percentile interval:*
$$\left[q_{\zeta/2}^*,\ q_{1-\zeta/2}^*\right].$$
- *Basic bootstrap interval:* centred to account for the possibility of a mismatch on the point estimate falling out of the confidence interval.
$$\left[2\hat{\phi}_{i,j,h} - q_{1-\zeta/2}^*,\ 2\hat{\phi}_{i,j,h} - q_{\zeta/2}^*\right].$$

However, the normality assumption of the innovation component does not always fit the analysis, so we could, in principle, use other methods to address such a problem, for example, **Non-Parametric residual-based iid bootstrap**, where we replace step (1) from previous algorithm with:

Given var Redisuals:

$$\hat{u_1}, ..., \hat{u_T}$$

Recenter them using the mean and get the same, but:

$$\hat{u_1}^c, ..., \hat{u_T}^c$$

and then perform iid resampling (with replacement).

Intuition behind the Bootstrap methodology

Bootstrap methodology allows us to solve many different nasty situations where inference might not be possible, in particular, let us think of the following example where:

$$x = \{2, 3, 4, 5, 6\}$$

Consider then $\hat{\mu} = \frac{1}{T}\sum_{i=1}^{N} x_i$, and our goal is to understand the distribution of such an estimator. What should I do? I can simply bootstrap and verify how far I get from the sample mean.

in our case let us think of

$$T^{1/2}(\hat{\theta}_T - \theta_0) \;\; N(0, V_\theta),$$

so that we can standardize the estimator as

$$\Gamma_T \;:=\; T^{1/2} V_\theta^{-1/2}(\hat{\theta}_T - \theta_0) \;\; N(0, I).$$

To define $V_\theta^{-1/2}$, take the spectral decomposition

$$V_\theta = P D_\theta P',$$

where D_θ is the diagonal matrix of eigenvalues of V_θ, and P is an orthonormal matrix satisfying $P'P = I = P^{-1}P$.

Then the symmetric square root of V_θ is

$$V_\theta^{1/2} := P D_\theta^{1/2} P',$$

and consequently

$$V_\theta^{-1/2} = P D_\theta^{-1/2} P'.$$

If we have a consistent estimator of the asymptotic covariance matrix V_θ, it holds that $\hat{\Gamma}_T$ distributes as a Normal $N(0, I)$, which is the $\Phi_z(x)$. Now we can define the bootstrap analogue:

$$\hat{\Gamma}_T^* := T^{1/2}\, \hat{V}_\theta^{-1/2}\, (\hat{\theta}_T^* - \hat{\theta}_T).$$

However, we do not know to which distribution it converges. We recall that it is important to distinguish between the two distributions we are working with:

- **Standard Probability Distribution on the data**
- **Synthetic Distribution induced by Bootstrap**

To get an understanding of the large sample distribution of our bootstrap estimate, we rely on conditioning on the data, so we let $G_T^*(x) = P(\hat{\Gamma}_T \leq x)$, where $G_T^*(x)$ is unknown. Using Bootstrap algorithms we can always build the empirical bootstrap cdf:

$$\hat{G}_{T:N}^*(x) = \frac{1}{N}\sum_{n=1}^{N} \mathbf{1}\Big(\hat{\Gamma}_{T:n}^* \leq x\Big) \qquad (\mathbf{1}(\cdot) \text{ indicator function}).$$

Glivenko-Cantelli Theorem: ensures that as $N \to \infty$ (with T fixed in this argument),

$$\sup_{x \in \mathbb{R}} \left| \hat{G}^*_{T,N}(x) - G^*_T(x) \right| \xrightarrow{\text{a.s.}} 0.$$

This means that we can approximate the unknown $G^*_T(x)$ arbitrarily well by using sufficiently many bootstrap replications of the estimator.

However, the real question is whether $G^*_T(x)$ is "close" to $\Phi_z(x)$, the cdf of $\hat{\Gamma}_T$.
We say that the bootstrap is *consistent* if, for every $\varepsilon > 0$,

$$\lim_{T \to \infty} \Pr\left(\sup_{x \in \mathbb{R}} |G^*_T(x) - \Phi_z(x)| > \varepsilon \right) = 0,$$

or equivalently,

$$\sup_{x \in \mathbb{R}} |G^*_T(x) - \Phi_z(x)| \xrightarrow{p} 0 \qquad \text{as } T \to \infty.$$

This is a *uniform convergence* result.

Under **Bootstrap Consistency**, we can claim that the bootstrap estimator, in the bootstrap world, has the same distribution as the sample estimator on the original data. SO we say that the convergence of the bootstrap estimator occurs in the bootstrap world conditional on the data.

Once we prove bootstrap consistency, we can safely build confidence intervals.

Chapter 6

Proxy SVAR

The **Proxy SVAR** approach arises from the interest in studying the identification of a set of shocks k such that $k < M$, where M is the number of variables we have in our SVAR model. This allows us to explore the *Partial identification logic*. Let us imagine we have the following situation:

$$\begin{pmatrix} u_{1,t} \\ u_{2,t} \\ u_{3,t} \end{pmatrix} = \begin{pmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{pmatrix} \begin{pmatrix} \varepsilon_t^{FU} \\ \varepsilon_t^{MU} \\ \varepsilon_t^{y} \end{pmatrix} \qquad \begin{matrix} \text{financial unc. shock} \\ \text{macro uncer. shock} \\ \text{real econ. activity shock} \end{matrix}$$

$$= (b_1 : b_2 : b_3) \begin{pmatrix} \varepsilon_t^{FU} \\ \varepsilon_t^{MU} \\ \varepsilon_t^{y} \end{pmatrix}$$

$$= \underbrace{\begin{pmatrix} b_{11} \\ b_{21} \\ b_{31} \end{pmatrix}}_{b_1} \varepsilon_t^{FU} + \underbrace{\begin{pmatrix} b_{12} \\ b_{22} \\ b_{32} \end{pmatrix}}_{b_2} \varepsilon_t^{MU} + \underbrace{\begin{pmatrix} b_{13} \\ b_{23} \\ b_{33} \end{pmatrix}}_{b_3} \varepsilon_t^{y}.$$

end, we are interested in studying only the financial uncertainty shock, we can use **Proxy-SVAR** approach by relying on a Z_t instrument variable which should satisfy the following two conditions:

- **Relevant**: $Cov(Z_t, \varepsilon_t^{FU}) = |\phi| >> 0$
- **Exogeneity**: the instrument is uncorrelated with the other two shocks.

We can therefore write everything as:

$$Z_t = \phi\, \varepsilon_t^{FU} + \underbrace{0 \times \varepsilon_t^{MU}}_{\text{exogeneity}} + \underbrace{0 \times \varepsilon_t^{y}}_{\text{exogeneity}} + \sigma_\omega \underbrace{\omega_t^\circ}_{\omega_t}$$

$$= \phi \varepsilon_{1,t} + \omega_t$$

where ω_t° is a normalized *measurement error* independent of ε_t, and $\sigma_\omega > 0$. Using this fact, we can postmultiply by the innovations and obtain:

$$Z_t u_t' = \phi \varepsilon_t^{FU} u_t' + \sigma_\omega \omega_t u_t'$$

Now, by taking the expectation, we can come up with the following:

$$E[Z_t u_t'] = \phi b_1$$

the right-hand side can be estimated through its natural estimator $\hat{\Sigma}_{u,Z} = \frac{1}{T}\sum_{t=1}^{T} \hat{u}_t Z_t$. Under stationary processes and *fairly general conditions* on the process that generates $(u_t', Z_t)'$ one has

$$\hat{\gamma}_T \xrightarrow{p} \gamma_0\,, \qquad T^{1/2}(\hat{\gamma}_T - \gamma_0) \xrightarrow{d} N(0, V_\gamma)$$

and **this result holds regardless of the strength of the proxies**.

Chapter 7

Appendix

Solution of a DSGE model an example

In this section we show how to end up with a unique and multiple solutions of a small and simple DSGE model. We start using the **Cagan's RE model** so we consider the rational expectations model of the form:

$$W_t = \alpha E_t[W_{t+1}] + u_t, \qquad \alpha > 0,$$

where u_t is a fundamental shock satisfying $E_t[u_{t+1}] = 0$. *Any stochastic process* $\{W_t\}$ *that satisfies the equation is a valid RE solution.* For example, $W_t = u_t$ is a solution since $E_t u_{t+1} = 0$.

In general, however, there are many possible solutions; in fact, the model is contaminated by the indeterminacy of some components, in the sense that there might be different stochastic processes satisfying the equation. Recall that the solution does not necessarily need to be stable; we could also have non-stationary solutions. To study them, use the identity

$$W_{t+1} = E_t[W_{t+1}] + \eta_{t+1},$$

where the forecast error $\eta_{t+1} = W_{t+1} - E_t[W_{t+1}]$ is a martingale difference sequence. Substituting this into the model yields

$$W_t = \alpha(W_{t+1} - \eta_{t+1}) + u_t,$$

which rearranges to

$$W_{t+1} = \frac{1}{\alpha} W_t + \eta_{t+1} - \frac{1}{\alpha} u_t.$$

The forecast error can be linearly projected onto the fundamental shock:

$$\eta_{t+1} = \text{Proj}(\eta_{t+1} \mid u_{t+1}) + s_{t+1} = \tau u_{t+1} + s_{t+1},$$

where τ is arbitrary and s_{t+1} is an orthogonal martingale difference sequence (a sunspot shock). Substituting gives

$$W_{t+1} = \frac{1}{\alpha} W_t + \tau u_{t+1} + s_{t+1} - \frac{1}{\alpha} u_t.$$

Ignoring the sunspot term for a moment, this expression is an ARMA(1,1) representation:

$$\left(1 - \frac{1}{\alpha} L\right) W_{t+1} = \left(\tau - \frac{1}{\alpha} L\right) u_{t+1} + s_{t+1}.$$

Stability depends only on the autoregressive root:

$$\left|\frac{1}{\alpha}\right| < 1 \qquad \Longleftrightarrow \qquad \alpha > 1.$$

Hence, for $\alpha > 1$ there are infinitely many stable (asymptotically stationary) solutions, because both τ and the sunspot process s_t are unrestricted.

If $s_{t+1} = 0$, the solution reduces to

$$\left(1 - \frac{1}{\alpha}L\right) W_{t+1} = \left(\tau - \frac{1}{\alpha}L\right) u_{t+1},$$

which is still indeterminate unless $\tau = 1$. When $\tau = 1$, the two polynomials share a root and cancel, giving the **Minimum State Variable solution**

$$W_{t+1} = u_{t+1}.$$

If $0 < \alpha < 1$, then $|1/\alpha| > 1$ and the model generates explosive paths. Again, for arbitrary τ there are multiple explosive solutions, while the special case $\tau = 1$ yields root cancellation and the unique stable (MSV) solution

$$W_{t+1} = u_{t+1}.$$

Companion Form Representation

Consider the VAR(p) model

$$y_t = A_1 y_{t-1} + A_2 y_{t-2} + \cdots + A_p y_{t-p} + u_t, \qquad u_t \sim (0, \Sigma_u), \tag{7.1}$$

where y_t is an $n \times 1$ vector of endogenous variables and $A_1, \ldots, A_p$ are $n \times n$ coefficient matrices. It is often convenient to rewrite this system as a first–order VAR in an expanded state vector. This is known as the *companion form*.

State Vector

Define the $np \times 1$ companion state vector

$$Y_t = \begin{pmatrix} y_t \\ y_{t-1} \\ \vdots \\ y_{t-p+1} \end{pmatrix}. \tag{7.2}$$

Companion Matrix

We can rewrite the VAR(p) as the first–order VAR:

$$Y_t = \mathcal{A}\, Y_{t-1} + \mathcal{U}_t, \tag{7.3}$$

where the *companion matrix* $\mathcal{A}$ is the $np \times np$ block matrix

$$\mathcal{A} = \begin{pmatrix} A_1 & A_2 & \cdots & A_{p-1} & A_p \\ I_n & 0 & \cdots & 0 & 0 \\ 0 & I_n & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & I_n & 0 \end{pmatrix}, \tag{7.4}$$

and the expanded error vector is

$$\mathcal{U}_t = \begin{pmatrix} u_t \\ 0 \\ \vdots \\ 0 \end{pmatrix}. \tag{7.5}$$

Interpretation and Use

The companion form is central for:

- deriving the Wold representation;
- studying stability through the eigenvalues of $\mathcal{A}$;
- computing impulse response functions (IRFs);
- expressing the VAR as a linear state–space system.

A VAR(p) is stable if and only if all eigenvalues of the companion matrix $\mathcal{A}$ lie strictly inside the unit circle. In that case, we obtain the moving–average representation

$$y_t = \sum_{j=0}^{\infty} \Phi_j \, u_{t-j}, \tag{7.6}$$

with coefficients determined recursively by $\Phi_0 = I_n$ and

$$\Phi_j = A_1\Phi_{j-1} + A_2\Phi_{j-2} + \cdots + A_p\Phi_{j-p}. \tag{7.7}$$

7.1 On The Weak White Noise

From the classical VAR model we can get $u_t = (u_{1,t}, ..., u_{M,t}9$ disturbances and we impose white noise on them therefore:

- $E[u_t] = 0_{M\times 1}$
- $E[u_t u_t'] = \Sigma_u < \infty$
- $E[u_t u_{t-k}'] = 0_{M\times M}, \;\; k = 1, 2, ...$

However we recall that Σ_u is the unconditional covariance matrix which symmetric and positive definite. So *we can relax the white noise up to a point*, by considering the minimal requirement of uncorrelation over time of the innovations, therefore we assume the u_t as forecast errors:

$$u_t = W_t - E[W_t|\mathcal{I}_{\sqcup -\infty}]$$

The MDS implies then that autocorrelation is equal to zero for the innovations, in fact, by the law of iterated expectations we have:

$$E\big(u_t u_{t-k}'\big) = E(E(u_t \mid \mathcal{I}_{t-k}))$$

$$= E\big(E(u_t \mid \mathcal{I}_{t-k})\, u_{t-k}'\big) = E\big(0_{M\times 1}\, u_{t-k}'\big) = 0_{M\times M}.$$

We can also strengthen this white noise condition by assuming **Strong** White noise (iid) so we impose then that $u_t|\mathcal{I}_{\sqcup -\infty} \sim N(0_{M\times 1}, \Sigma_u)$

However this assumption does not hold in macro/finance since we generally get fat tails. This can be shown using the ARCH (1) model which shows us:

$$U_t = \sigma_t \eta_t, \sigma_t \perp \eta_t \sim iidN(0,1)/T(g)$$

rewriting the model:

$$\sigma_t^2 = \alpha_0 + \alpha_1 u_{t-1}^2, \quad \alpha_0 > 0, \alpha_1 \in (0,1)$$

it is possible to show that:

$$E[u_t^2] = k$$

$$E[u_t u'_{t-k}] = 0$$

so the ARCH(1) is uncorrelated and is unconditionally homeostatic, now given this we can add $(+/-)\sigma_t^2$ to $u_t^2 : u_t^2 = \sigma_t^2 + \underbrace{sigma_t^2(\eta_t^2 - 1)}_{v_t}$ therefore:

$$u_t^2 = \alpha_0 + \alpha_1 u_{t-1}^2 + v_t$$

So we have an AR(1), which is a correlated process.

7.2 Instrumental Variable

In this section, we are interested in studying the instrumental variable approach to solve problems that could undermine the validity of econometric analysis under some conditions. Given the y_t a scalar dependent variable and x_t a scalar independent variable such that $w_t := (y_t, x_t)'$ and $\{w_t\}$ is stationary and ergodic:

- $\frac{1}{T}\sum_{i=1}^{T} w_t \xrightarrow{p} E[w_t]$
- $\frac{1}{T}\sum_{i=1}^{T} w_t w'_{t-q} \xrightarrow{p} E[w_t w'_{t-q}]$, for $q = 0, 1, 2...$

assuming $u_t \sim WN(0, \sigma^2)$ can specify the classic linear model where:

$$y_t = \beta x_t + u_t$$

Our Parameter β can be estimate using what follows:

$$\hat{\beta}_T^{OLS} = \beta + \frac{\frac{1}{T}\sum_{i=1}^{T} x_t u_t}{\frac{1}{T}\sum_{i=1}^{T} x_t^2}$$

Given the aforementioned formula we typically assume $E[x_t u_t] = 0$, so $\frac{1}{T}\sum_{i=1}^{T} x_t u_t \xrightarrow{p} E[x_t u_t] = 0$ Hence we assume $\beta_T^{OLS} \xrightarrow{p} \beta_0$.

Whenever we estimate an OLS regression, we implicitly assume that causality runs from the regressor x_t to the dependent variable y_t. In other words, to explain y_t we do not require a structural relationship in which x_t is itself determined by y_t. If such reverse causality were present, the bidirectional link between the two variables would generate endogeneity, thereby biasing the OLS estimators and invalidating standard inference. Intuitively, this assumption reflects the idea that our estimator is also incorporating the effects of our regressor with the error term therefore we are not able to distinguish the pure relation relation between x_t adn y_t.

$$\hat{\beta} \xrightarrow{p} \beta + \frac{\gamma}{\Sigma_{xx}}$$

To address this problem, we rely on an instrument that allows us to obtain a "purified" estimate of the regressor x_t, denoted by $\hat{x}_t^F$. This fitted value does not incorporate the component of u_t that contaminates x_t. We then regress y_t on the first-stage prediction $\hat{x}_t^F$, which is by construction uncorrelated with the structural error term.

For the estimator to be consistent, the instrument must satisfy the following conditions:

1. $E[z_t u_t] = 0$ (Exogeneity)
2. $E[x_t z_t] \neq 0$ (Relevance)

If both conditions hold, then z_t is a valid instrument. We obtain that:

We have:

$$\hat{\beta}_T^{IV} := \frac{\sum_{t=1}^T y_t z_t}{\sum_{t=1}^T x_t z_t} \quad = \quad \text{under DGP}$$

$$= \frac{\sum_{t=1}^T (\beta_0 x_t + u_t) z_t}{\sum_{t=1}^T x_t z_t} = \beta_0 \frac{\sum_{t=1}^T x_t z_t}{\sum_{t=1}^T x_t z_t} + \frac{\sum_{t=1}^T z_t u_t}{\sum_{t=1}^T x_t z_t}$$

$$= \beta_0 + \frac{\frac{1}{T}\sum_{t=1}^T z_t u_t}{\frac{1}{T}\sum_{t=1}^T x_t z_t}.$$

Exploiting asymptotics, we get:

$$\hat{\beta}^{IV} \xrightarrow{p} \beta$$

We define the IV regression as follows:
$\begin{cases} y_t = \beta x_t + u_t \ , & \text{Second Stage} \\ x_t = \Pi z_t + v_t \ , & \text{First Stage} \end{cases}$

Given that we recall that we have:

- **Endogeneity** is measured by $E[x_t u_t] = E[(\Pi z_t + v_t) u_t] =$ and comes only from the "relation" between the x_t and the error term.
- **Relevance** is given by $E[x_t z_t] = \Pi \sigma_z^2$ if it is close to zero we have **weak instrument** which causes deviation from gaussian behavior as $T \to \infty$

Example where $H_p : E(x_t u_t) = 0$ does *not* hold

We consider the estimation of the relationship between inflation and slack (Phillips-type curve):

$$\pi_t = \beta \tilde{x}_t + u_t,$$

where π_t is the inflation rate and $\tilde{x}_t$ a measure of the output gap.

Suppose instead that the true DGP belongs to the AD–AS structural model (aggregate demand + aggregate supply) with exogenous monetary policy:

$$\tilde{x}_t = \gamma \tilde{x}_{t-1} - \delta(R_t - \pi_t) + \sigma_{\tilde{x}} \varepsilon_{\tilde{x},t}, \tag{7.8}$$

$$\pi_t = \kappa \tilde{x}_t + \sigma_\pi \varepsilon_{\pi,t}. \tag{7.9}$$

We assume

$$\varepsilon_{\tilde{x},t} \sim WN(0,1), \qquad \varepsilon_{\pi,t} \sim WN(0,1),$$

with

$$\varepsilon_{\tilde{x},t} \perp \varepsilon_{\pi,t}, \qquad R_t \perp \varepsilon_{\tilde{x},t}, \qquad R_t \perp \varepsilon_{\pi,t}.$$

Then:

$$\tilde{x}_t \text{ causes } \pi_t \quad \text{through the Phillips curve.}$$

However, also:

$$\pi_t \text{ causes } \tilde{x}_t \quad \text{through the AD (IS) curve.}$$

Therefore:

$$E(\tilde{x}_t u_t) \neq 0.$$

Why? Because the omitted error term in the reduced-form Phillips curve contains shocks that also drive $\tilde{x}_t$, generating reverse causality and feedback from π_t to $\tilde{x}_t$.

To compute $E(\tilde{x}_t u_t)$, we first derive the reduced form associated with the structural AD–AS model.

Write the system in matrix form:

$$\underbrace{\begin{pmatrix} 1 & -\delta \\ -\kappa & 1 \end{pmatrix}}_{R_0} \begin{pmatrix} \tilde{x}_t \\ \pi_t \end{pmatrix} = \underbrace{\begin{pmatrix} \gamma & 0 \\ 0 & 0 \end{pmatrix}}_{R_1} \begin{pmatrix} \tilde{x}_{t-1} \\ \pi_{t-1} \end{pmatrix} + \underbrace{\begin{pmatrix} -\delta \\ 0 \end{pmatrix}}_{F} R_t + \underbrace{\begin{pmatrix} \sigma_{\tilde{x}} & 0 \\ 0 & \sigma_{\pi} \end{pmatrix}}_{\Sigma_\varepsilon} \begin{pmatrix} \varepsilon_{\tilde{x},t} \\ \varepsilon_{\pi,t} \end{pmatrix}.$$

This representation is the starting point for deriving the reduced form and computing the implied correlation between $\tilde{x}_t$ and the composite error term u_t in the mis-specified Phillips regression. From this, we can recover the associated reduced form, which is given by:

$$\begin{pmatrix} \tilde{x}_t \\ \pi_t \end{pmatrix} = \underbrace{\begin{pmatrix} \pi_{1,1} & 0 \\ \pi_{2,1} & 0 \end{pmatrix}}_{\Pi_1} \begin{pmatrix} \tilde{x}_{t-1} \\ \pi_{t-1} \end{pmatrix} + \underbrace{\begin{pmatrix} \lambda_{1,1} \\ \lambda_{2,1} \end{pmatrix}}_{\Lambda} R_t + \begin{pmatrix} \nu_{x,t} \\ \nu_{\pi,t} \end{pmatrix}.$$

Here, the parameters are a nonlinear function of the structural parameters. Now, considering the first equation, we see that both x_{t-1} and R_t are predetermined, so they are not correlated with the error term $v_{x,t}$, but they are correlated with x_t. Therefore, they are valid instruments. We can decide to use both instruments jointly.